

Chapter

Probability

1. Assertion (A): A die is thrown once and the probability of getting an even number is $\frac{3}{2}$.

Reason (R): The sample space for even numbers on a die is {2, 4, 6}.

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- (a) A is true, R is false.
- (b) A is false, R is true.
- (c) Both A and R are true.
- (d) Both A and R are false.

2. Event A: The sun will rise from east tomorrow.

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Event B: It will rain on Monday.

Event C: February month has 29 days in a leap year. Which of the above event(s) has probability equal to 1?

- (a) all events A, B and C
- (b) both events A and B
- (c) both events B and C
- (d) both events A and C

3. A letter is chosen at random from all the letters of the English alphabets. The probability that the letter chosen is a vowel, is :

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- (a) $\frac{4}{26}$
- (b) $\frac{5}{26}$
- (c) $\frac{21}{26}$
- (d) $\frac{5}{24}$

4. The probability of getting a number divisible by 3 in throwing a dice is :

- (a) $\frac{1}{6}$
- (b) $\frac{1}{3}$
- (c) $\frac{1}{2}$
- (d) $\frac{2}{3}$

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46. A bag contains 25 cards, numbered through 1 to 25. A card is drawn at random. What is the probability that the number on the card drawn is:

(a) multiple of 5

(b) a perfect square

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(c) a prime number?

47. A bag contains cards numbered 1 to 49. Find the probability that the number on the drawn card is :

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(i) an odd number

(ii) a multiple of 5

(iii) Even prime

48. A bag contains 5 white, 2 red and 3 black balls. A ball is drawn at random.

What is the probability that the ball drawn is a red ball? Main 2022 Sem II

49. A letter of the word SECONDARY is selected at random. What is the probability that the letter selected is a vowel?

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Solution

1.

Assertion (A):

A die is thrown once and the probability of getting an even number is $\frac{3}{2}$.

- Possible outcomes on a die: {1, 2, 3, 4, 5, 6}
- Even numbers: {2, 4, 6} → 3 outcomes
- Probability of getting an even number = $\frac{3}{6} = \frac{1}{2}$, **not** $\frac{3}{2}$

✗ Assertion (A) is false

Reason (R):

The sample space for even numbers on a die is {2, 4, 6}.

- This statement is correct.

✓ Reason (R) is true

✓ **Correct Answer:**

(b) A is false, R is true.

2.

Event A:

The sun will rise from east tomorrow.

This is a **sure event** (certain to happen).

👉 Probability = 1

Event B:

It will rain on Monday.

This is an **uncertain event** (may or may not happen).

👉 Probability $\neq$ 1

Event C:

February month has 29 days in a leap year.

This is always true for a leap year.

👉 Probability = 1

Events with probability equal to 1:

✔ Event A

✔ Event C

✔ **Correct answer:**

(d) both events A and C

3.

The vowels are:

A, E, I, O, U → 5 vowels

Probability of choosing a vowel:

$$\text{Probability} = \frac{\text{Number of favourable outcomes}}{\text{Total outcomes}} = \frac{5}{26}$$

✔ **Correct answer:**

(b) $\frac{5}{26}$

4.

When a die is thrown, the possible outcomes are:

$$\{1, 2, 3, 4, 5, 6\}$$

Numbers divisible by 3 are:

$$\{3, 6\}$$

So,

- Number of favourable outcomes = 2
- Total possible outcomes = 6

$$\text{Probability} = \frac{2}{6} = \frac{1}{3}$$

✔ **Correct answer:**

(b) $\frac{1}{3}$

46.

(a) Probability that the number is a multiple of 5

Multiples of 5 between 1 and 25 are:

5, 10, 15, 20, 25

Number of favourable outcomes = 5

$$P(\text{multiple of 5}) = \frac{5}{25} = \frac{1}{5}$$

(b) Probability that the number is a perfect square

Perfect squares between 1 and 25 are:

1, 4, 9, 16, 25

Number of favourable outcomes = 5

$$P(\text{perfect square}) = \frac{5}{25} = \frac{1}{5}$$

(c) Probability that the number is a prime number

Prime numbers between 1 and 25 are:

2, 3, 5, 7, 11, 13, 17, 19, 23

Number of favourable outcomes = 9

$$P(\text{prime number}) = \frac{9}{25}$$

 **Final Answers:**

- (a) $\frac{1}{5}$
- (b) $\frac{1}{5}$
- (c) $\frac{9}{25}$



47.

(i) Probability that the number is an odd number

From 1 to 49, the odd numbers are:

$$1, 3, 5, \dots, 49$$

Total odd numbers = 25

$$P(\text{odd number}) = \frac{25}{49}$$

(ii) Probability that the number is a multiple of 5

Multiples of 5 between 1 and 49 are:

$$5, 10, 15, 20, 25, 30, 35, 40, 45$$

Number of multiples of 5 = 9

$$P(\text{multiple of 5}) = \frac{9}{49}$$

(iii) Probability that the number is an even prime

The only even prime number is:

$$2$$

Number of favourable outcomes = 1

$$P(\text{even prime}) = \frac{1}{49}$$

 **Final Answers:**

- (i) $\frac{25}{49}$
- (ii) $\frac{9}{49}$
- (iii) $\frac{1}{49}$

48.

There are total $5 + 2 + 3 = 10$ ball in bag.

Thus $n(S) = 10$

No. of red ball i.e. favorable outcomes,

$$n(E) = 2$$

$$P(E) = \frac{n(E)}{n(S)} = \frac{2}{10} = 0.2$$

49.

The word SECONDARY has the following letters:

S, E, C, O, N, D, A, R, Y

Step 1: Total number of letters

There are **9 letters** in the word SECONDARY.

Step 2: Identify the vowels

Vowels are: A, E, I, O, U

Vowels present in SECONDARY are:

E, O, A

Number of vowels = 3

Step 3: Find the probability

$$P(\text{vowel}) = \frac{\text{Number of vowels}}{\text{Total number of letters}} = \frac{3}{9} = \frac{1}{3}$$

 **Answer:**

The probability that the letter selected is a vowel is $\frac{1}{3}$.